Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2013-0402-DS-2-RV-102	Issued	2013-09-09

UT Thickness Data Sheet - 2-RV-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 2-RV-102, located within the ATU-100. The inspection was conducted as part of the 3-year regenerator overhead and reflux survey detailed in HIS-2013-0402.

Kevin Doucette, acting as lead UT technician (SNT-TC-1A UT Level II) for Halward Inspection Services Ltd., performed the measurements and data recording for this line.

Technical Basis and Procedures

Thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 2 and HIS-UT-002 Rev 1. These procedures govern the site NDE requirements for the CUI program, including the use of inspection ports and insulation removal.

The measurement methodology and equipment calibration were executed in accordance with ASTM E797.

Component Specifications

The subject line, 2-RV-102, is a regenerator overhead circuit constructed from A106 Gr.B. The line has an outer diameter of 60.3 mm and a nominal thickness of 5.54 mm.

The design limits for this component are 345 kPa(g) and 121 °C. The minimum allowable thickness (t-min) is 2.79 mm.

Evaluation Criteria

Acceptance is based on the retirement thickness, defined as the t-min of 2.79 mm. Any reading below this value is subject to an NCR.

Corrosion rates are determined by the larger of the long-term rate or the short-term rate over the latest pair of readings at least 3 years apart. Remaining life is calculated as (t - t-min) / rate.

Inspection Interval and Classification

Line 2-RV-102 is designated as Class 1 due to its service as sour gas regenerator overhead. The class maximum for thickness and external visual inspection is 5 / 5 years.

The regenerator overhead circuit is currently maintained on a 3-year thickness interval. Under the CUI program, insulated lines in the CUI range require profile RT or insulation windows every 5 years.

Thickness readings, measured 2013-09-09

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2010-09-13	6.78	6.49	0.063	0.077	38.1	2016-09-09	
3-RV-101-02	7.62	3.56	2010-09-13	6.59	6.44	0.065	0.075	38.3	2016-09-09	
3-RV-101-03	7.62	3.56	2010-09-13	6.16	5.80	0.098	0.127	17.6	2016-09-09	
3-RV-101-04	7.62	3.56	2010-09-13	6.43	6.13	0.083	0.089	29.0	2016-09-09	
3-RV-101-05	7.62	3.56	2010-09-13	6.63	6.48	0.068	0.065	43.0	2016-09-09	
3-RV-101-06	7.62	3.56	2010-09-13	6.43	6.11	0.085	0.100	25.4	2016-09-09	
3-RV-101-07	7.62	3.56	2010-09-13	6.20	5.73	0.157	0.157	13.8	2016-09-09	
3-RV-101-08	7.62	3.56	2010-09-13	6.71	6.36	0.117	0.117	23.9	2016-09-09	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2010-09-13	4.55	4.49	0.059	0.057	28.6	2016-09-09	
2-RV-102-02	5.54	2.79	2010-09-13	4.60	4.29	0.074	0.064	20.2	2016-09-09	
2-RV-102-03	5.54	2.79	2010-09-13	4.84	4.68	0.046	0.055	34.2	2016-09-09	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-04	5.54	2.79	2010-09-13	4.84	4.81	0.044	0.052	38.9	2016-09-09	
2-RV-102-05	5.54	2.79	2010-09-13	5.04	4.96	0.027	0.027	>50	2016-09-09	
2-RV-102-06	5.54	2.79	2010-09-13	4.82	4.55	0.090	0.090	19.5	2016-09-09	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2006-02-24	5.48	5.39	0.005	0.012	>50	2023-09-09	
3-RF-101-02	5.49	2.41	2006-02-24	5.37	5.34	0.006	0.004	>50	2023-09-09	
3-RF-101-03	5.49	2.41	2006-02-24	5.45	5.45	0.000	0.000	>50	2023-09-09	
3-RF-101-04	5.49	2.41	2006-02-24	5.33	5.33	0.007	0.000	>50	2023-09-09	
3-RF-101-05	5.49	2.41	2006-02-24	5.40	5.26	0.011	0.019	>50	2023-09-09	
3-RF-101-06	5.49	2.41	2006-02-24	5.40	5.25	0.015	0.020	>50	2023-09-09	
3-RF-101-07	5.49	2.41	2006-02-24	5.35	5.43	-0.001	-0.011	>50	2023-09-09	
3-RF-101-09	5.49	2.41	—	—	5.12	—	—	—	2023-09-09	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2006-02-24	3.85	3.71	0.012	0.019	>50	2023-09-09	
2-RF-102-02	3.91	2.16	2006-02-24	3.88	3.65	0.013	0.031	48.8	2023-09-09	
2-RF-102-03	3.91	2.16	2006-02-24	3.80	3.76	0.010	0.005	>50	2023-09-09	
2-RF-102-04	3.91	2.16	2006-02-24	3.81	3.80	0.004	0.001	>50	2023-09-09	
2-RF-102-05	3.91	2.16	—	—	3.69	—	—	—	2023-09-09	
2-RF-102-06	3.91	2.16	—	—	3.71	—	—	—	2023-09-09	
2-RF-102-07	3.91	2.16	2006-02-24	3.85	3.82	0.008	0.004	>50	2023-09-09	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2013-09-09	_____